

Homework Assignment 4 - Coding Part Write-up

Networks and Markets

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Part 4: Page Rank and Social Networks

1 Question 7

The two other simple test cases we used are:

Extra Graph 1 The first graph is two cycles connected by a single edge (i.e., an infinite loop).

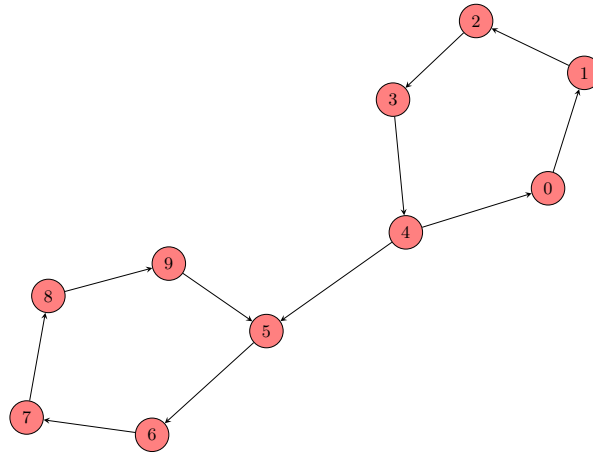


Figure 1: Extra Graph 1

Extra Graph 2 The second graph is a bipartite graph with 5 nodes on each side, where each node on one side is connected to all nodes on the other side.

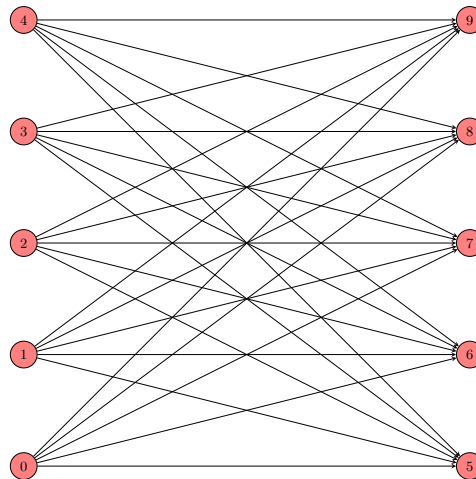


Figure 2: Extra Graph 2

2 Question 8

- (c) Running PageRank on the given graph, we get that nodes with high ranks typically have higher degrees, and nodes with low ranks typically have lower degrees.

The top 5 nodes with the highest page rank had the following degrees: 547, 1045, 792, 347, 755 (where top rank had 547 neighbors, and 5th had 755).

The bottom 5 nodes with the lowest page rank had the following degrees: 1, 1, 1, 1, 1.

- (d) Intuitively, the page rank of a node is the probability that a random walker will be at that node at a given time.

If a node has a high rank, then starting a random walk from many nodes it is likely to arrive at that node.

Since the graph is equivalent to an undirected one, it means that a high rank means that random walks from it might reach many nodes.

If we want to quantify influence over people using it, it seems that the scaled page rank is a good measure.